OSHE – Module 5

Quick Revision – Main Points

Q1. Preventive Measures for Safety in Construction Industry

1. **Mobile Plant Safety** – train drivers; dedicated routes for vehicles; improved visibility systems; regular vehicle inspection
2. **Working at Height** – permit-to-work system; task risk assessment; safety harnesses and nets; display permits and warnings
3. **Ladder Safety** – inspect before use; tie/foot securely; extend 1 m above platform; maintain 1-out-for-4-up angle
4. **Scaffolding Safety** – erected/dismantled by certified personnel only; inspect before first use, after alterations, after storm, every 7 days; handrails + toe boards mandatory
5. **Mobile Scaffold Tower** – erected on firm level ground; barricade below; access through internal ladders only; handrails provided
6. **Quarry Face Safety** – edge protection using strapping and aluminium poles; ratcheted barriers; safety harnesses and lanyards; train workers on harness use
7. **Safety Nets** – install during roof/elevated work; use harnesses at secure anchorage points; protect workers where continuous attachment is impractical

Q2. Safety and Health in Wastewater Treatment Plants

1. **Chemical Hazards** – chlorinated solvents, PCBs, pesticides, heavy metals, dioxins; cause eye/throat irritation, organ damage, cancer
2. **Biological Hazards** – disease-causing microorganisms in sewage/sludge; cause nausea, diarrhoea, flu, waterborne infections; new workers most susceptible
3. **Inhalation Hazards** – aerosols, vapours, gases from aeration tanks and dewatering; cause respiratory disorders and gastrointestinal exposure
4. **Skin Contact Hazards** – chemical absorption, dermatitis, infections through cuts, needle-stick injuries during bar-screen cleaning
5. **Engineering Controls** – adequate ventilation; enclosed treatment systems; reduce aerosol dispersion at design stage
6. **Administrative Controls** – regular safety training; restrict hazardous area access; monitor worker exposure
7. **PPE** – safety gloves, protective clothing, safety boots, respiratory protection, eye/face protection
8. **Medical Surveillance** – regular health check-ups; monitor occupational illnesses; maintain health records

Q3. Roles and Responsibilities of Managers in Construction Industry

1. **Safety Policy Formulation** – establish written safety policy; define objectives; allocate responsibilities; monitor compliance
2. **Safety Communication** – communicate site safety plans, procedures, and performance feedback; encourage free information flow
3. **Safety Training** – induction training for new workers; special training for high-risk jobs (scaffold, crane); periodic refresher programs
4. **Accident Investigation** – investigate causes; recommend corrective measures; maintain accident records and safety performance data
5. **Site Planning** – participate in pre-site planning; identify hazards before work starts; allocate safety resources
6. **Safe Work Methods** – develop safe systems for hazardous operations; involve workers; conduct risk assessments
7. **Safety Committees** – establish and support safety committees; act as technical advisers; review safety performance
8. **Contractor Supervision** – ensure contractor compliance with regulations; select subcontractors based on safety record; conduct audits
9. **PPE and Resources** – ensure PPE availability; allocate sufficient resources for safety and emergency preparedness
10. **Emergency Preparedness** – develop emergency and security plans; organize response teams; conduct regular drills

Q4. Handling of Chemicals in Laboratories

1. **Read Labels and MSDS** – understand hazards, storage requirements, and first-aid measures before handling any chemical
2. **Use PPE** – safety goggles, chemical-resistant gloves, lab coat/apron, safety shoes, face shields where needed
3. **Proper Storage** – store by compatibility; keep flammables away from heat; cool, dry, ventilated areas; containers tightly closed and labelled
4. **Avoid Direct Contact** – never touch chemicals with bare hands; use lab tools for transfer; wash hands thoroughly after handling
5. **Prevent Inhalation** – work in well-ventilated areas; use fume hoods; wear respirators for airborne hazardous chemicals
6. **Follow Chemical Safety Rules** – use prescribed procedures only; never mix unknown chemicals; use only required quantity
7. **Emergency Measures** – eye contact – rinse with water immediately; skin – wash with water, remove contaminated clothing; inhalation – move to fresh air; seek medical attention
8. **Safety Signage and Labelling** – display hazard symbols and warning signs; post emergency contact info in lab
9. **Fire and Explosion Prevention** – keep flammables from ignition sources; maintain fire extinguishers; follow spill-control procedures

Q5. Potential Hazards at Construction Sites

1. **Mobile Plant and Vehicle Hazards** – collisions from dumper trucks, forklifts, front-loading shovels; workers struck by moving equipment; traffic accidents
2. **Fall Hazards (Working at Heights)** – falling from roofs, ladders, scaffolds, platforms; leading cause of serious injuries and deaths
3. **Scaffold Hazards** – falls from unprotected edges; scaffold collapse; falling tools and materials; improper access injuries
4. **Quarry Face Hazards** – falls from edges; rock falls; loss of balance during drilling or loading shot holes
5. **Slips, Trips and Falls** – uneven surfaces, poor housekeeping, obstructions, wet/slippery floors; significant percentage of all injuries
6. **Dust Hazards** – cement and construction dust causes respiratory irritation, breathing difficulties, COPD, reduced visibility
7. **Noise and Vibration Hazards** – excessive noise from machinery leads to hearing loss, deafness, fatigue, and discomfort
8. **Fire and Explosion Hazards** – flammable materials, solvents, combustible dust; improper chemical storage; cause burns, property damage, fatalities
9. **Hazardous Material Exposure** – chemicals, solvents, toxic dusts; cause eye irritation, skin disorders, poisoning, respiratory problems
10. **Confined Space Hazards** – oxygen deficiency, poisonous gases, dust accumulation, fire risk, suffocation; sudden filling by liquids or solids
11. **Electrical Hazards** – electric shocks, burns, electrocution from faulty wiring, exposed conductors, or improper isolation

Q6. Occupational Health Hazards in an Epoxy Manufacturing Unit

1. **Chemical Hazards** – epoxy resins, hardeners, curing agents, solvents; cause skin irritation, chemical burns, allergic reactions, toxic effects
2. **Respiratory Hazards** – vapours, fumes, dust from mixing/processing; cause throat irritation, asthma-like symptoms, chronic respiratory disorders, lung damage
3. **Skin Hazards** – frequent contact causes dermatitis, skin sensitization, redness, itching, chemical burns; most common occupational problem in epoxy industries
4. **Eye Hazards** – chemical splashes cause eye irritation, conjunctivitis, corneal damage, temporary or permanent vision impairment
5. **Fire and Explosion Hazards** – flammable solvents; vapour accumulation; fire accidents; burns and smoke inhalation
6. **Physical Hazards** – excessive noise from machinery; heat from chemical reactions; slips from chemical spills; moving equipment injuries
7. **Ergonomic Hazards** – manual handling of raw materials; causes back pain, musculoskeletal disorders, fatigue, repetitive strain injuries
8. **Psychological Hazards** – stress from handling hazardous chemicals; shift-work fatigue; mental strain from continuous process monitoring

Q7. Roles and Responsibilities of Workers and Managers in Safety Programs

A. Workers

1. **Follow Safety Rules** – obey all safety regulations and instructions; use safe work methods; follow workplace policies
2. **Use PPE** – wear helmets, gloves, safety shoes, goggles, harnesses; maintain PPE in good condition
3. **Participate in Toolbox Briefings** – attend toolbox talks; discuss hazards with supervisors; clarify safe work doubts
4. **Conduct Safety Checks** – inspect workplace before operations; identify and report unsafe conditions immediately
5. **Report Hazards and Accidents** – inform supervisors about unsafe conditions; report near misses and defective equipment; cooperate during investigations
6. **Protect Fellow Workers** – work responsibly to avoid endangering others; encourage safe practices among coworkers

B. Managers

1. **Develop Safety Policies** – establish written safety policy; define objectives and standards; allocate responsibilities at all levels
2. **Organize Safety Training** – conduct induction training for new staff; periodic refresher programs; ensure understanding of hazards
3. **Communicate Safety Information** – share site safety plans; promote two-way communication; share accident lessons learned
4. **Investigate Accidents** – analyze root causes; recommend preventive measures; maintain accident records and safety reports
5. **Participate in Site Planning** – identify hazards before work starts; ensure adequate safety arrangements in project plans
6. **Establish Safe Work Systems** – develop safe methods for hazardous operations; ensure risk assessments are conducted and followed
7. **Act as Safety Advisors** – assist safety committees; provide technical guidance; promote strong safety culture across the organization
8. **Ensure PPE and Resources** – provide necessary PPE and safety equipment; ensure workers receive adequate supervision and support

Q8. Problems Associated with Water and Wastewater Treatment Plants

1. **Accident Hazards** – slips/falls on wet floors; falls from ladders; drowning; machinery injuries; electric shock; suffocation in confined spaces; fire and explosion
2. **Physical Hazards** – high noise from pumps/motors causes hearing impairment; adverse weather (heat/cold stress); UV radiation from disinfection processes
3. **Chemical Hazards** – chlorine, hydrofluoric acid, sodium hypochlorite, ozone, coagulants; cause eye irritation, skin burns, lung oedema, toxic poisoning
4. **Biological Hazards** – disease-causing microorganisms in wastewater/sludge; cause nausea, diarrhoea, indigestion, flu, infectious diseases

5. **Inhalation Hazards** – aerosols, vapours, gases, airborne droplets from aeration/dewatering; cause respiratory disorders and long-term health problems
6. **Skin Contact Hazards** – irritation, chemical absorption, dermatitis, infections through wounds, needle-stick injuries from screenings
7. **Ergonomic Problems** – awkward postures during inspection; heavy lifting; repetitive tasks; cause back pain and musculoskeletal disorders
8. **Psychosocial Problems** – stress from noise, humidity, and odours; shift work and overtime; workload pressure; adapting to computer-based systems

Q9. Safety Measures to Prevent Accidents at Water and Wastewater Treatment Plants

1. **Use PPE** – safety helmets, goggles, face shields, gloves, respirators, non-skid safety shoes, protective clothing
2. **Chemical Safety** – check supply connection points regularly; display warning signs; read and follow MSDS; store chemicals separately by property
3. **Engineering Controls** – adequate ventilation; exhaust ventilation for toxic gases; machine guards; enclosed systems where possible
4. **Confined Space Safety** – check air quality before entry; test for oxygen deficiency and toxic gases; ventilate; use safety harnesses; standby person outside
5. **Electrical Safety** – inspect equipment before use; repair defects immediately; use qualified electricians; lockout/tagout during maintenance
6. **Prevent Slips and Falls** – non-skid footwear; keep floors clean and dry; inspect and secure ladders; install guardrails around elevated areas
7. **Biological Hazard Control** – avoid direct contact with wastewater; maintain personal hygiene; wash hands after work; cover wounds; use protective gloves
8. **Safe Manual Handling** – proper lifting techniques; use lifting aids for heavy loads; avoid awkward postures; seek assistance for bulky equipment
9. **Noise Protection** – wear appropriate ear protection; monitor noise levels; limit exposure to machinery noise
10. **Training and Medical Surveillance** – regular safety training; educate on chemical and biological hazards; periodic medical examinations; monitor worker health
11. **Emergency Preparedness** – first-aid facilities; emergency showers and eye-wash stations; established emergency response procedures; regular drills

Q10. Safety and Health in RMC Plant

1. **Dust Hazards** – cement dust from handling/mixing causes respiratory irritation, COPD; limit at source; good housekeeping; wear respiratory PPE
2. **Noise and Vibration Hazards** – mixers, crushers, conveyors cause hearing loss and deafness; wear earplugs; monitor workers' hearing regularly
3. **Fire and Explosion Hazards** – combustible materials, solvents, oils; store separately in cool dry places; fire protection systems; regular fire drills

4. **Hazardous Material Exposure** – cement, chemicals, solvents cause skin/eye irritation and burns; wear PPE (gloves, goggles, boots, hard hats); flush with water if exposed
5. **Material Transport Hazards** – truck/loader/forklift accidents; use inspected and maintained vehicles; trained drivers; separate pedestrian and vehicle routes; no mobile phone while driving
6. **Confined Space Hazards** – oxygen deficiency, toxic gases, dust, fire risk; permit-to-work system; atmospheric testing before entry; proper ventilation; standby person outside
7. **Electrical Hazards** – electric shocks and burns; turn off isolation switches before maintenance; verify isolation; use lockout/tagout with individual keys

Q11. Problems Associated with Cement Plants

1. **Quarrying Hazards** – falling rocks, vehicle accidents, falls near quarry edges, dust from drilling and blasting operations
2. **Crushing Hazards** – direct contact with crushing machinery, injuries from moving parts, falling objects, traffic near crushers
3. **Clinker Production Hazards** – burns from hot clinker, alkali burns from cement kiln dust (CKD), chemical exposure, high-temperature fire hazards
4. **Milling Process Hazards** – hot material contact, dust exposure, mechanical injuries during maintenance, blockage removal risks in raw/coal mills
5. **Material Transport Hazards** – vehicle collisions, traffic accidents, falling loads, injuries during loading and unloading operations
6. **Confined Space Hazards** – oxygen deficiency, poisonous gases/vapours, dust accumulation, entrapment by solids or liquids, suffocation
7. **Hazardous Material Exposure** – eye irritation, skin disorders, chemical burns, inhalation poisoning from cement and process chemicals
8. **Electrical Hazards** – electric shock, burns, equipment malfunction, accidents due to improper isolation of electrical systems
9. **Dust Hazards** – inhalation of cement dust causes respiratory irritation, COPD, reduced workplace visibility
10. **Noise and Vibration Hazards** – crushers and mills cause hearing impairment, deafness due to prolonged exposure, worker fatigue and discomfort
11. **Fire and Explosion Hazards** – combustible materials, dust accumulation, flammable solvents and oils, toxic smoke and vapours during fires
12. **Slips, Trips and Falls** – uneven quarry surfaces, poor housekeeping, obstacles in work areas, wet and slippery floors